Heart Disease Prediction Using Random Forest

A Classical Machine Learning Study on the UCI Cleveland Heart Disease Dataset

Abstract

This study investigates Random Forest for binary heart disease prediction using the UCI Cleveland Heart Disease dataset. The workflow includes data inspection, missing-value handling, binary target transformation, stratified train-test splitting, Random Forest training, classification evaluation, feature-importance analysis, confusion-matrix analysis, and prediction error analysis. The primary model uses 500 trees and random_state=42. On the reported 80/20 split, the full-feature model achieved approximately 90% test accuracy. An exploratory ablation removing age produced approximately 92% accuracy. Because the dataset is small and the main result uses one holdout split, these results should be treated as an educational experiment rather than evidence of clinical readiness.

1. Research Question

Can a Random Forest classifier trained on the UCI Cleveland Heart Disease dataset achieve strong binary classification performance while providing interpretable feature-importance information and revealing patterns in its prediction errors?

2. Research Motivation

Heart disease prediction provides a useful benchmark for studying supervised classification because the Cleveland dataset combines demographic, categorical, and continuous clinical attributes. Recent work continues to compare Random Forest with classical and deep learning approaches while emphasizing interpretability. This project focuses on a lightweight ensemble baseline and studies not only predictive performance but also feature importance and model errors.

3. Research Objectives

O1. Develop a reproducible Random Forest classifier for binary heart disease prediction.

O2. Evaluate the classifier using accuracy, precision, recall, F1-score, support, and a confusion matrix.

O3. Rank predictors using Random Forest feature-importance scores.

O4. Identify misclassified test observations and compare important-feature distributions between correct and incorrect predictions.

O5. Investigate whether removing age changes test-set performance.

4. Related Work and Literature Gap

The UCI repository reports 303 instances and 13 commonly used predictive features for the Cleveland dataset. The original target represents absence or presence of angiographic heart disease. Breiman's Random Forest method combines randomized decision trees and also provides variable-importance measures.

Research on the Cleveland dataset includes conventional machine learning, feature selection, ensemble methods, explainability methods, and increasingly complex deep learning systems. Recent studies have examined Random Forest alongside Logistic Regression, SVM, KNN, XGBoost and neural models, while other work has combined heart-disease prediction with LIME, permutation importance or SHAP.

The gap addressed by this project is educational and methodological: demonstrate a compact, reproducible Random Forest baseline and then inspect its behavior through feature importance and error analysis rather than treating accuracy as the final answer.

5. Experimental Method

Dataset: Cleveland processed data from the UCI Heart Disease repository.

Preprocessing: '?' markers are replaced with NaN, values are converted to numeric form, missing values are filled using median imputation, and the original multi-level target is converted to binary classification.

Split: 80% training and 20% testing using stratification and random_state=42.

Model: RandomForestClassifier with n_estimators=500 and random_state=42.

Evaluation: classification report and confusion matrix.

Interpretability: feature_importances_ is ranked and visualized.

Error analysis: test samples are labeled as correct or incorrect predictions, and the six most important features are compared using KDE plots.

Ablation: age is removed in a separate experiment and test performance is compared with the full-feature model.

6. Results

The reported full-feature experiment achieved approximately 90% accuracy on the 61-sample test set, with approximately 0.90 F1-score for both classes.

When age was removed, the reported accuracy increased to approximately 92%, with approximately 0.92 F1-score for both classes. This is an exploratory result showing that removing a feature can sometimes improve a particular model on a particular split.

The final notebook calculates Random Forest feature_importances_ and ranks the predictors. The exact top-three features and scores should be taken directly from the final published run rather than copied from external studies.

The confusion matrix and error-analysis plots provide additional information about which samples were classified incorrectly and whether important feature distributions differ between correct and incorrect predictions.

7. Discussion

The results show that Random Forest can achieve strong performance on this small benchmark without a complex neural architecture. The age-ablation result is particularly useful because it demonstrates why feature selection should be tested empirically.

However, the 90% to 92% change should not be interpreted as evidence that age is generally harmful. The experiment uses one holdout split, so sampling variation may contribute to the difference. A stronger study would use repeated stratified cross-validation and tune hyperparameters using training folds only.

Feature importance should also be interpreted as model reliance rather than causation. Gini importance can be influenced by feature characteristics and correlated predictors. Permutation importance or SHAP would provide complementary interpretability.

The model is not a clinical diagnostic system. Clinical use would require external validation, calibration, prospective evaluation, fairness analysis, and medical and regulatory oversight.

8. Limitations and Future Work

The dataset is small and the main result is based on one 80/20 split. The current notebook also performs median imputation before the split; a production-quality implementation should fit preprocessing only on training data to prevent leakage. Hyperparameter tuning, repeated cross-validation, probability calibration, threshold analysis, external validation, and stronger explainability methods are not yet included.

Future work should include StratifiedKFold cross-validation, GridSearchCV or RandomizedSearchCV, leakage-safe preprocessing pipelines, permutation importance/SHAP, calibration, and external validation.

9. Contribution and Conclusion

This project demonstrates a complete classical machine learning workflow for heart disease classification: data preparation, reproducible splitting, ensemble training, multi-metric evaluation, feature-importance analysis, confusion-matrix analysis, and direct investigation of prediction errors.

The reported full-feature model achieved approximately 90% test accuracy, while the exploratory age-ablation experiment achieved approximately 92%. These results are encouraging for an educational benchmark but are not sufficient to establish clinical usefulness.

The main lesson is that machine learning does not end with a high accuracy score. A stronger workflow asks what the model learned, which features it relied on, where it failed, and whether those observations remain stable under rigorous validation.

References

[1] Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart Disease. UCI Machine Learning Repository. DOI: 10.24432/C52P4X.

[2] Breiman, L. (2001). Random Forests. Machine Learning, 45, 5–32.

[3] UCI Machine Learning Repository. Heart Disease dataset documentation.

[4] Dual explainability framework for heart disease prediction using LIME and permutation feature importance. Discover Applied Sciences, 2025/2026.

[5] Empirical analysis of predicting heart disease using diverse datasets and classification procedures of machine learning. 2025.